Robot Arm Load Analyzer - Full Roadmap

Robot Arm Load Analyzer — Full Implementation Roadmap

3D, full Denavit–Hartenberg kinematics, dynamic load analysis, realistic rendering

A direct, honest note before anything else: what you're describing now — full 3D, proper DH kinematics, *dynamic* (not just static) load analysis, and realistic rigged rendering — is a genuine multi-week robotics project. It's the kind of scope a graduate robotics course spends a whole semester on. That's not a reason not to do it; it's a reason to **phase it properly**, so the midterm has a real, working, honestly-scoped deliverable, and the rest of the vision has a clear path instead of being crammed into days and coming out broken. This document is that phased plan — now updated with specific, concrete lessons pulled from the two closest existing tools, so the same mistakes aren't repeated from scratch.

Lessons From Existing Tools (Prior Art)

No existing tool does all of this at once — confirmed by direct research into the two closest ones. That split itself is informative: each tool independently chose to narrow its scope along almost the exact same boundary this roadmap already uses (kinematics-only vs. dynamics-only, 3D vs. full physics), which is a strong independent validation that the Phase 1 → 2 → 3 split here is the right way to tame this problem, not an arbitrary one.

ComPU-Tools Robot Kinematics Simulator — full 3D, DH-based, 2–6 DOF, real-time, but *zero* dynamics. Specific techniques worth adopting directly:

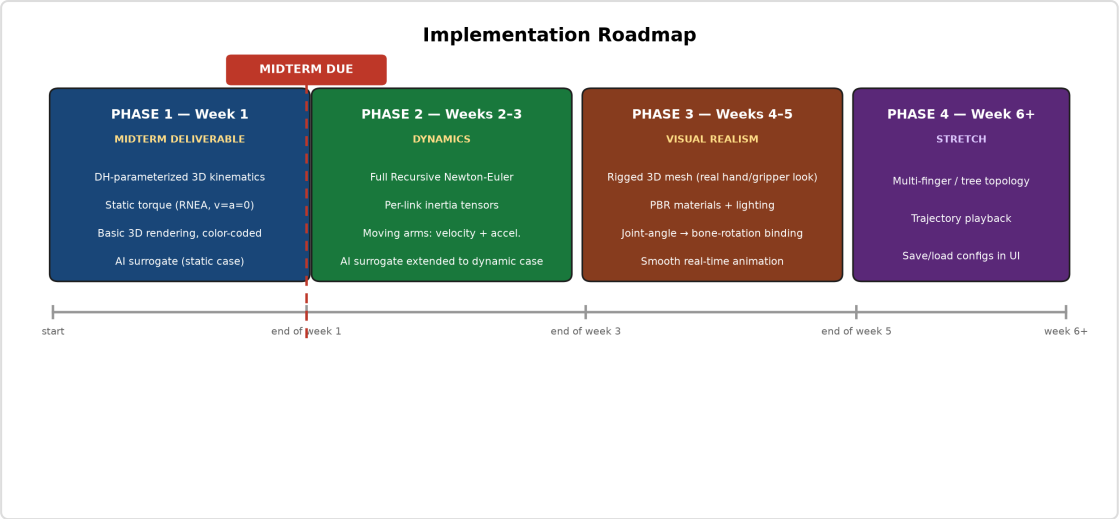
- **Angle winding.** Their iterative solver normalizes every revolute joint angle to `[−180°, 180°]` at every step — without this, accumulated rotations can drift to arbitrarily large values over time. **Adopt this in Phase 1**: normalize `theta_i` every time it's updated, not just when it's first set.
- **Global unit consistency.** One unit system (mm/m/in for length, deg/rad for angle) enforced across every input, DH table entry, and result display — inconsistent units across joints is a real, easy-to-hit bug class once there's more than one length/angle field in the UI. **Adopt in Phase 1.**
- **3D camera pivot, not world-origin orbiting.** Orbiting the camera around world `(0,0,0)` makes inspecting a specific joint frustrating once the arm extends away from the origin. Their fix: an adjustable view-pivot point at the canvas center that orbit/zoom/pan all operate relative to, persisted across sessions. **Adopt in Phase 1's UI** — this is a small addition with an outsized usability payoff.
- **Scope discipline, confirmed.** They explicitly do not render CAD meshes/motor housings and explicitly do not compute dynamics, by design — "focuses purely on kinematics." This is the same

Phase 1/Phase 3 split this roadmap already makes, arrived at independently by someone else solving a neighboring problem.

NovaSolver Newton-Euler Dynamics Calculator — real Newton-Euler dynamics (M/C/G terms), real-time, but locked to a 2-DOF *planar* arm, no 3D, no AI. Specific techniques worth adopting directly:

- **Trajectory smoothing matters, concretely.** Stepping a joint angle directly (instant jump from one target to another) makes velocity and acceleration spike from zero to maximum instantly, producing unrealistic torque spikes and simulated motor overload. Their fix: interpolate motion with a **cubic-spline profile** that guarantees zero velocity at both trajectory endpoints. **Adopt in Phase 2**: whenever the tool animates between two poses (not just a single static snapshot), use a smooth interpolation profile instead of a step change, or the dynamic torque numbers will be dominated by an artifact of the UI, not the arm.
- **RNEA over symbolic Lagrangian, confirmed.** They explicitly chose the recursive Newton-Euler formulation over symbolic (Lagrangian) dynamics because symbolic expressions get too large to re-evaluate every frame at interactive speed. This directly validates Phase 2's algorithm choice — it's not just textbook-standard, it's the one a real interactive tool had to use for performance.
- **Decompose the torque, don't just show the total.** Rather than a single number, they separately compute and display the inertia term, the Coriolis/centrifugal term, and the gravity term — as a live breakdown bar chart — so an engineer can see *why* a joint is loaded, not just *how much*. **Adopt in Phase 2's UI**: since RNEA naturally computes these pieces internally anyway, surfacing them separately costs almost nothing extra and is a genuinely more useful readout than a single torque number.
- **Document your physical assumptions explicitly.** They assume every link is a uniform slender rod (`I = mL²/12`, center of mass at the midpoint) and say so plainly, rather than quietly baking it in. This project already plans to support arbitrary per-link inertia tensors (a real upgrade over this assumption) — but the lesson stands: **whatever simplifying assumption is in place at any given phase should be stated in the README, not left implicit.**
- **State what's deliberately out of scope.** No automatic motor-overload warnings, no gearbox/friction losses — both explicitly left to the user, with a stated recommendation to apply a 1.5–2.0× safety factor manually. Worth carrying the same honesty into this project's own README once Phase 2 ships: say plainly that the computed torque is the *ideal* theoretical value, not a motor-selection guarantee.

Roadmap Overview



Implementation roadmap: four phases from the midterm-scoped static/DH foundation through full dynamics to realistic rendering and stretch goals.

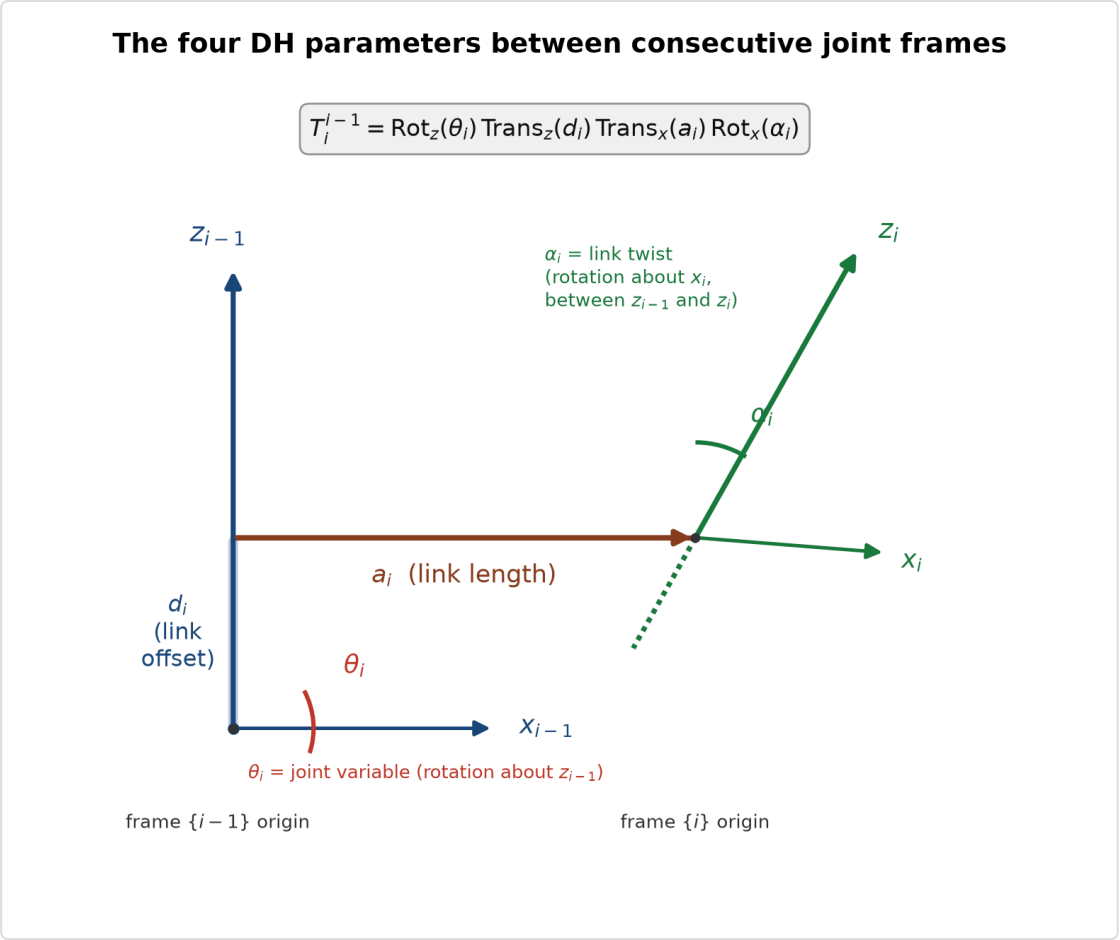
Phase	Timeframe	Delivers
1	Week 1 (midterm)	Real 3D kinematics via full DH parameters, static load analysis, working 3D view with visible joints
2	Weeks 2–3	Full dynamic analysis (moving arms, not just held poses)
3	Weeks 4–5	Realistic rendering — the arm/hand actually *looks* real, not primitive shapes
4	Week 6+	Stretch: multi-finger hands, trajectory playback, saved configs

Each phase below includes what it is, the math behind it, and what changes in the code.

Phase 1 — DH Kinematics + Static Load Analysis (Week 1, Midterm)

Why DH parameters, not simple trig

The earlier plan computed joint positions with basic 2D trigonometry — fine for a flat arm, but it doesn't generalize to a real 3D arm where joints can twist on different axes (a shoulder that rotates sideways *and* an elbow that hinges up-down, for instance). **Denavit–Hartenberg (DH) parameters** are the standard, general way to describe *any* serial-link robot's geometry, 3D included, using just four numbers per joint.



The four DH parameters between consecutive joint frames, and the transformation matrix they define.

For joint *i*, the four parameters are:

Parameter	Meaning
`a_i`	**link length** — distance between joint axes `z_{i-1}` and `z_i`, along their common perpendicular
`alpha_i`	**link twist** — angle between `z_{i-1}` and `z_i`, about that common perpendicular
`d_i`	**link offset** — distance along `z_{i-1}` to where the common perpendicular meets it
`theta_i`	**joint angle** — the one you actually *control*; rotation about `z_{i-1}`

Each joint's transform relative to the previous one is a single 4×4 matrix:

```
T_i^(i-1) = Rot_z(theta_i) * Trans_z(d_i) * Trans_x(a_i) * Rot_x(alpha_i)
```

Chain them together and you get the full pose (position *and* orientation, in 3D) of every joint and the end-effector, for *any* arm geometry — not just planar ones:

```
T_n^0 = T_1^0 * T_2^1 * T_3^2 * ... * T_n^(n-1)
```

This is the real generality upgrade. An arm is now fully described by a table of `(a, alpha, d)` triples (one row per joint) plus joint limits — drop in any serial arm's DH table and the whole kinematics chain works unmodified.

> **Convention warning, genuinely worth flagging:** there are two common DH conventions — "standard/classic" (used above) and "modified" (Craig's textbook). They order the matrix multiplications differently and are a classic source of silent, hard-to-debug sign/axis errors. Pick one, write down which one in your README, and don't mix references that use the other.

Static load analysis, reframed correctly for 3D

The earlier "sum of moments" shortcut only works cleanly in 2D. The correct, general approach — and the one that sets up Phase 2 cleanly — is the **Jacobian transpose method**:

```
tau = J^T * F_ext
```

where `J` is the manipulator Jacobian (built directly from the DH transforms — it relates joint velocities to end-effector velocity) and `F_ext` is the external wrench (force + torque) applied at the end-effector — here, the payload's weight. This is the textbook-correct way to map an end-effector load to joint torques for *any* 3D arm, and it's a natural stepping stone to full dynamics: **static analysis is just the dynamic equations (Phase 2) with zero velocity and zero acceleration.** Same algorithm, simpler special case — that framing is exactly how real robotics code is usually structured, so you're not throwing away Phase 1 work when you get to Phase 2.

What ships at the end of Phase 1

- DH-table-driven 3D forward kinematics (any serial arm, not just planar)
- Static joint torques via Jacobian transpose (gravity + payload)
- A working 3D view — basic geometry (cylinders for links, spheres for joints) is enough here; the "looks like a real hand" visual polish is Phase 3, deliberately deferred so Phase 1 stays achievable in a week
- The AI surrogate, trained on this static case (same idea as before: sweep random poses/payloads through the exact DH+Jacobian calculation, train a small model on the result)
- **Angle normalization** (`theta_i` wrapped to `[-180°, 180°]` on every update) and **one consistent unit system** end to end — both cheap to build in from day one, both real bug classes if skipped (see "Lessons From Existing Tools" above)
- **3D view pivot at canvas center**, not world origin, for camera orbit/pan/zoom — small addition, large usability payoff once the arm isn't sitting at `(0,0,0)`

Phase 2 — Full Dynamic Analysis (Weeks 2–3)

Static analysis answers "what torque holds this pose." Dynamic analysis answers the harder, more useful question: "what torque does this joint need *right now*, given the arm is also moving" — accounting for velocity, acceleration, and how every link's inertia reacts to that motion. This is what a real robot's controller actually needs, and it's what makes the tool able to analyze an arm *swinging* under load, not just posed and frozen.

The Recursive Newton-Euler Algorithm (RNEA)

This is the standard algorithm (every robotics textbook — Craig, Spong, Siciliano — has a chapter on it). Two passes over the chain, per timestep:

Forward pass (base → tip) — propagate motion outward, link by link:

```
omega_i     = R * omega_(i-1)  +  theta_dot_i * z_i             (angular
velocity)
omega_dot_i = R * omega_dot_(i-1) + R*omega_(i-1) x (theta_dot_i*z_i) +
theta_ddot_i*z_i   (angular accel.)
a_i         = (rigid-body acceleration of link i's center of mass,
               computed from omega_i, omega_dot_i, and the previous link's
acceleration)
```

Backward pass (tip → base) — propagate force/torque inward, link by link:

```
F_i   = m_i * a_i                                   (Newton: force needed to
accelerate this link)
N_i   = I_i * omega_dot_i  +  omega_i x (I_i * omega_i)   (Euler: torque needed,
including gyroscopic/Coriolis effects)
tau_i = (project the accumulated joint force/torque onto this joint's axis z_i)
```

Setting every velocity and acceleration to zero collapses this exactly to Phase 1's static case — which is why Phase 1 isn't wasted work.

What this requires that Phase 1 didn't

- **A 3×3 inertia tensor per link** (not just a point mass) — how a link resists *rotational* acceleration, which depends on its shape/mass distribution, not just its weight
- **Joint velocities and accelerations as inputs** — either set directly ("what if this joint is spinning at X rad/s"), or computed from a user-defined trajectory (a sequence of poses over time)
- Noticeably more careful bookkeeping — this pass genuinely has more moving parts than Phase 1, which is exactly why it's its own multi-week phase rather than a bullet point

What ships at the end of Phase 2

- Full RNEA: the arm can be analyzed *while moving*, not just held still

- The AI surrogate extended to the higher-dimensional dynamic input (pose + velocity + acceleration + payload → torque) — a meaningfully harder regression problem than Phase 1's, likely needs a larger network and more training data
- UI: ability to specify a velocity/acceleration (or scrub through a simple trajectory) and watch the torque numbers respond in real time, not just change with static pose
- **Cubic-spline trajectory interpolation** for any pose-to-pose animation, instead of stepping joint angles directly — avoids fake torque spikes that are a UI artifact, not real physics (see "Lessons From Existing Tools")
- **Torque readout decomposed into its three physical causes** (inertia term, Coriolis/centrifugal term, gravity term), not just a single combined number — turns the readout from "here's a torque" into "here's *why* this joint is loaded"
- A stated, documented list of what the physics does *not* model (gearbox friction, motor-rated-torque checking) — honest about being the ideal theoretical torque, not a motor-selection guarantee

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Phase 3 — Realistic 3D Rendering (Weeks 4–5)

This is the "looks like a real hand" part, and it's worth separating cleanly from the physics/AI work above — it's a genuinely different skill set (3D art/rigging pipeline), and bolting it onto Phase 1 would have blown the midterm deadline for no good reason, since the physics and AI parts don't care what the arm looks like.

Two realistic paths, in order of effort

Path A — procedural but polished (do this first). Keep the link/joint geometry procedural (cylinders, capsules, spheres — generated from code, same as Phase 1), but invest in **Qt Quick 3D's PBR (physically-based rendering) materials and lighting**: brushed-metal or matte-plastic material on the links, proper ambient + directional lighting, subtle shadows. This alone gets you most of the way to "looks like a real mechanical arm" without touching 3D modeling software at all — materials and lighting do more visual work than people expect.

Path B — an actual rigged mesh (if you want the literal hand/gripper look). Source or build a rigged 3D model (a robotic gripper or an articulated hand — e.g. a free, appropriately-licensed rigged model, or a simple one built in Blender), import it as glTF into Qt Quick 3D, and **drive its bone rotations from your computed joint angles every frame** instead of your own procedural geometry. This is a real, well-established technique (it's how every 3D game animates characters), but it adds genuine pipeline work:

- Matching the model's bone hierarchy to your DH joint chain (they need to correspond 1:1, or you need a mapping between them)
- Aligning coordinate systems — your DH frames and the model's bone-local frames are very unlikely to agree out of the box, and a sign/axis mismatch here is the single most common bug in this kind of work
- Keeping the rig's rest pose and your "all joint angles = 0" pose consistent, so `theta_i` actually means the same thing to the physics and to the mesh

Recommendation: do Path A thoroughly first. It's lower-risk, ships faster, and the earlier UI mockup already assumed this level of polish. Treat Path B as what you reach for only if Path A is done early and you have real time left — it's genuinely a "nice to have that's also a time sink," not a core requirement to prove the project works.

What ships at the end of Phase 3

- A 3D arm that looks intentional and real — proper materials, lighting, and (if Path B) an actual rigged mesh — not placeholder primitives
- Smooth real-time animation as poses/loads change, color-coding carried over from Phase 1

Phase 4 — Stretch Goals (Week 6+)

Not required for the core project to be complete and impressive — listed so there's an obvious "here's what's next," which is also exactly what makes the project something *others* could pick up and extend:

- **Multi-finger / tree-structured kinematics** — if "real hand" specifically means a multi-fingered gripper rather than a single arm, this is where it belongs: each finger is its own DH chain, all sharing one base/palm frame. A genuinely different (harder) topology than a single serial chain, so it's correctly scoped as its own phase, not folded into Phase 1 or 3.
- **Trajectory playback** — record/scrub through a sequence of poses over time, watching dynamic torque change live (this is the single best demo of why Phase 2's dynamics work mattered).
- **Save/load arm configurations directly from the UI**, instead of hand-editing a config file.

What Changes in "What You'll Need"

Everything from the original plan still applies (Qt 6, C++17 + CMake, Eigen, Python+PyTorch for training). Added for this expanded scope:

Need	For
Qt Quick 3D	3D rendering (Phases 1 and 3)
A DH-table data structure/format (e.g., a simple JSON schema: array of `{a, alpha, d, joint_type}`)	Making the arm genuinely swappable, per the original generality goal
Inertia tensor data per link	Phase 2 dynamics
(Phase 3, Path B only) a glTF-compatible 3D modeling tool (e.g., Blender, free) and/or a sourced rigged model	Realistic mesh rendering

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The One-Sentence Version

Week 1 proves the hard engineering is correct (real 3D kinematics, real static physics) and ships something real for the midterm. **Weeks 2–3** make it genuinely dynamic. **Weeks 4–5** make it look like it's worth showing someone. **Week 6+** is bonus.

Why This Is Still Worth Building

To close the loop on the prior-art research: ComPU-Tools proves the 3D/DH/real-time-interactive side is a solved, approachable problem. NovaSolver proves the dynamics side is too — for a 2-DOF planar arm. Neither one is the other, and neither has a trained AI surrogate sitting on top. This project is the union of both halves, in 3D, with the AI layer — a combination that, as far as this research could find, nobody has shipped as one tool. That's the actual pitch.